

ASHUTOSH RAJPUT

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📁 GeeksForGeeks 📁 InterviewBit

EDUCATION

Indian Institue of Technology Hyderabad

M.Tech - Computer Science & Engineering

Hyderabad, India

Jul. 2023 – Jun. 2025

College Of Engineering Roorkee

B.Tech - Computer Science & Engineering

Roorkee, India

Jul. 2018 – Jun. 2022

TECHNICAL SKILLS & TOOLS

- Languages — Java, C/C++, Python, SQL *Elementary Proficiency
- Core CS Fundamentals — Object-Oriented Programming, Operating Systems, Algorithms, Data Structures
- Development Tools — IntelliJ Idea, Android Studio, GitHub, Perforce*, VS Code
- Networking Tools — Wireshark, Docker*

WORK EXPERIENCE

Engineer

Jul. 2025 - Present

Samsung R&D Institute, Noida

- Analyzing SELinux audit logs and system reports to identify permission denials and security-related issues across various services.
- Resolving identified issues by configuring SELinux policies, managing labels, and assigning appropriate permissions to ensure secure and seamless system operation.

PROJECTS

PIRSense: Activity Detection via PIR Sensor Data [\[GitHub\]](#)

Mar. 2025 - May. 2025

Domain: Deep Learning and IoT

- Engineered and compared multiple feature selection strategies (manual selection, information gain, PCA, and Feature-engine-based selection) to optimize model performance across 4 curated datasets.
- Designed and trained deep learning models (MLP & LSTM) on each dataset, evaluating them using metrics like accuracy, precision, recall, F1-score, and runtime.
- Conducted rigorous model benchmarking and analysis, demonstrating MLP's consistent performance across datasets, while identifying LSTM's challenges with specific classes and feature selection methods—validated using 5-fold cross-validation.
- Tech Stack: [Python, TensorFlow, Keras, scikit-learn, pandas, NumPy, Matplotlib, Seaborn, Feature-engine]

Secure Chat App [\[GitHub\]](#)

Mar. 2024 – Apr. 2024

Domain: Network Security

- Developed a secure P2P chat app using DTLS 1.2 over UDP with encryption, session resumption, mutual authentication, and Perfect Forward Secrecy (PFS)
- Simulated MITM downgrade and active attacks with forged certificates to validate security measures.
- Tech Stack: [C++, OpenSSL, DTLS, X.509 Certificates]

PUBLICATIONS

Denial of Service Attacks Targeting Layer 2 in 5G RAN

IEEE FNFw 2025

Domain: O-RAN Security

- Proposed and validated a UE-based detection technique that monitors MAC and RLC layer events (e.g., RACH and RRCSetup failures) to identify FBS activity early, enhancing network resilience
- Designed and implemented attack scenarios like RACH response and RRCSetupRequest drops on both physical and simulated testbeds, demonstrating the effectiveness of the detection algorithm in preventing resource drain.
- Suggested a PCI blacklisting mechanism for UEs to share suspected FBS instances, enabling decentralized, early detection and reducing reliance on operator infrastructure for mitigating threats
- Tech Stack: [O-RAN, Docker, USRP, OAI]

ACHIEVEMENTS

- Ranked #1 in Indian Institute Of Technology Hyderabad on GeeksForGeeks.
- Ranked #1 In Indian Institute Of Technology Hyderabad on InterviewBit.
- Ranked among the top 250 coders globally on GeeksforGeeks.
- Achieved a global rank of 62 in GFG Weekly-181 Contest 2024.
- Secured an All India Rank (AIR) of 428 in GATE 2023.